

INDUSTRIAL INTERNSHIP REPORT

on

AI & Automation Systems for Cable Manufacturing & Operations

Submitted To

Department of Computer Science & Engineering
School of Engineering & Technology

In partial fulfilment of the requirements for the degree of
Bachelor of Technology in Computer Science & Engineering

Student Name:	Girish Kumar Yadav
Roll Number:	CS-2341412
Batch / Session:	2023–2027 (4th Year / 7th Semester)
Section:	4CSE4
Host Organization:	KDI Power Private Limited, Delhi
Internship Duration:	29 June 2026 – 31 August 2026 (2 Months)
Certificate No.:	KDIP/INT/2026/AI-014

Candidate's Declaration

I, **Girish Kumar Yadav**, Roll No. **CS-2341412**, student of B.Tech 7th Semester, Section **4CSE4** (Session **2023–27**), Department of Computer Science and Engineering, hereby declare that the internship report titled "**AI & Automation Systems for Cable Manufacturing & Operations**" is an authentic record of my own work carried out during my two-month industrial internship from **29 June 2026 to 31 August 2026** at **KDI Power Private Limited**, New Delhi, under Certificate No. **KDIP/INT/2026/AI-014**.

All references and resources used in this report have been duly acknowledged. I affirm that this report has not been submitted elsewhere for the award of any other degree or diploma.

Signature: _____

Student Name: Girish Kumar Yadav

Roll Number: CS-2341412

Batch: 2023–27 (4CSE4)

Acknowledgement

I express my deepest gratitude to the management of **KDI Power Private Limited**, New Delhi, for granting me the opportunity to undergo a two-month industrial internship in Artificial Intelligence & Automation. I am particularly indebted to **Mr. Vijay Gautam**, Administrative Manager, for his continuous guidance, encouragement, and invaluable domain insights throughout the project lifecycle.

I also extend my sincere appreciation to the faculty members of the Department of Computer Science and Engineering for their constant support and academic foundation. Finally, I thank my family and peers for their unceasing motivation.

Signature: _____

Student Name: Girish Kumar Yadav

Roll Number: CS-2341412

1. Executive Summary & Company Profile

KDI Power Private Limited is an ISO 9001, ISO 14001, and ISO 45001 certified manufacturer of electrical wires and cables headquartered in New Delhi, India. The company manufactures Low Voltage (LT) Power Cables, Aerial Bunched Cables (ABC), XLPE Power Cables, Control Cables, House Wires, and Rubber Cables for DISCOM utilities, EPC contractors, and global distributors.

During the 2-month industrial internship (June 29 – Aug 31, 2026), I worked as an **AI & Automation Engineer** and developed five major production applications to automate sales, lead generation, tender tracking, document processing, and task management.

2. Technical Projects Portfolio

Project 1: KDI Power AI WhatsApp Assistant & Sales Dashboard

Built an automated WhatsApp bot using FastAPI, Meta WhatsApp Cloud API, Groq AI (Llama 3.3 70B), Supabase PostgreSQL (pgvector RAG), and a real-time glassmorphic single-page sales dashboard for analytics and price editing.

Project 2: KDI Lead Intelligence & B2B Lead Discovery Engine

Engineered a 3-tier lead discovery pipeline with Google Maps Playwright scraping, proxy rotation, Groq LLM scoring engine (0-100 scale), and automated formatted Excel exports (.xlsx) targeting cable distributors in India and South Africa.

Project 3: LinkedIn Tender & Government Scheme Tracker

Developed an automated scraper using Playwright CDP attached to Chrome profiles and Mistral AI summarization to track competitor updates (Polycab, KEI, Finolex, Havells) and government electrification orders (RDSS, DDUGJY).

Project 4: AI-Powered PDF Reconstructor & OCR Editor

Constructed a hybrid web editor with React 18 / Vite and FastAPI using MinerU layout detection, NVIDIA NIM VLM (Llama 3.2 90B Vision) for table/math formula extraction, and Playwright headless Chromium for true PDF reconstruction.

Project 5: Google Sheets & WhatsApp Task Automation Engine

Deployed a FastAPI webhook on Render connected to Google Sheets API v4 service accounts and WhatsApp Meta API for automated employee task reminders and real-time IST completion timestamping.

3. Summary Matrix of Internship Deliverables

#	Project	Tech Stack	Status
1	WhatsApp Sales Bot	FastAPI, Meta API, Groq, Supabase	Deployed
2	Lead Intelligence	Playwright, Groq, SQLite, Excel	Deployed
3	LinkedIn Tender Tracker	Playwright CDP, Mistral AI, Excel	Deployed
4	AI PDF Editor	React 18, Vite, FastAPI, MinerU, NIM	Deployed
5	Task Reminder Bot	FastAPI, Google Sheets API, Render	Deployed